



RiderNet

₹140.

That's what Karthik earned last Wednesday.

He planned for ₹700.

School fees were due Friday.

His bike EMI hits on the 5th.

The rain didn't care.

The app didn't care.

No insurance product cared.

We did.

He keeps the city fed.

The city never kept him safe.

Until RiderNet.

About This Project

This is not a typical README.

This is the story of **Karthik**, a delivery partner — and the system that should have protected him.

Every feature in RiderNet answers one question:

“Does this actually help the rider?”

Problem Statement

Delivery partners face:

- No income during disruptions
- No protection for lost wages
- No formal financial safety net

Existing Systems Fail

System	Limitation
Health Insurance	Covers injuries, not income loss
Vehicle Insurance	Covers damage, not earnings
Platforms	No downtime compensation
Loans	Increase debt

Our Solution — RiderNet

Parametric Income Protection for Gig Workers

Automatic payouts triggered by real-world events.

Smart Payout Triggers

Trigger	Condition	Payout
Rain	> 64.4mm	Full
Heat	> 45°C	50%
AQI	> 400	Partial
Flood	Govt confirmed	Full
Bandh	Verified	Full

Adversarial Defense & Anti-Spoofing Strategy

The Attack

500 fake riders → Fake GPS → Real payouts → System drained

Core Problem:

Parametric insurance pays on **data**, not **presence**

Our Multi-Layer Defense System

Layer 1 — GPS & Physics Validation

GPS can be faked. Physics cannot.

We detect:

- Teleportation (>5 km jump)
- Unrealistic speeds (>40 km/h in rain)
- No movement for long durations
- Abnormal altitude changes

3 violations → Claim blocked

Also:

- Mock GPS detection (`isFromMockProvider`)
 - Device fingerprint tracking
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Layer 2 — Behaviour Analysis

We compare:

Signal	Real Rider	Fraud
Past activity	Regular	None
Zone presence	Consistent	Random
Work pattern	Continuous	Trigger-only

New accounts → **50% payout cap**

Layer 3 — Claim Pattern Detection

- Real events → smooth claim distribution
- Fraud → sudden spike (e.g., 400 claims in minutes)

Action: Temporary hold + verification

Layer 4 — Multi-Signal Verification

We require **3 out of 5 signals**:

- GPS
- Cellular tower
- Order activity
- Weather data
- Historical presence

Fraud fails

Karthik passes → gets paid instantly

Layer 5 — Fraud Ring Detection

We build a **claim graph**:

- Nodes = claims
- Edges = shared data (device, IP, UPI)

Detect entire fraud networks, not individuals

Layer 6 — Fairness & Risk Scoring

Score	Action
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0–30	Instant payout
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31–60 Partial payout

61–80 Review

81–100 Block + appeal

Rider Protection

- Appeal system (video proof)
 - Rural flexibility (2/5 signals)
 - Trust boost after false flags
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Pricing Model

Weekly — aligned with rider income

₹29 + Risk Score Adjustment
Example: ₹74.50/week

Plans

Tier	Price	Coverage
Basic	₹29	₹500
Standard	₹49	₹750
Premium	₹79	₹900

AI/ML Engine

System	Tech	Role
Premium Engine	XGBoost	Pricing
Fraud Detection	Isolation Forest	Anomaly detection
Forecasting	Prophet	Risk prediction

Workflow

1. Rider registers
 2. Premium calculated
 3. Event detected
 4. Fraud checks run
 5. Payout processed
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App Features

- Multi-language UI
- 2-min onboarding
- Real-time alerts
- Earnings dashboard

Rainy Day Mode

UI adapts emotionally during disruptions

Tech Stack

Frontend

- React + Vite
- TailwindCSS

Backend

- FastAPI
- PostgreSQL

AI/ML

- XGBoost
- Isolation Forest
- Prophet

APIs

- OpenWeatherMap
 - OpenAQ
 - NewsAPI
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Status

Phase 1 — Ideation

- Problem validated
 - System designed
 - Prototype pending
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Team

Bannari Amman Institute of Technology, Erode

- Anusha S — Full Stack
 - Abinaya S — Backend & ML
 - Jeyashri M — UI/UX
 - Kamalika K — Data
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Final Thought

Friday morning.
School fees paid.
Family okay.

That's not a feature.
That's the goal.

RiderNet

Because no rider should lose income to the weather.
